

AMANATULLAH PANDU ZENKLINOV

South Jakarta, Indonesia

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INTRODUCTORY PROFILE



Statistical Data Scientist & AI/ML Engineer with a strong foundation in rigorous statistical modeling and applied machine learning. Expert in bridging theoretical concepts (Causal Inference, Bayesian Methods) with production-grade engineering (Python, RAG Systems, Cloud Deployment). Proven track record in optimizing algorithms for scalability and deploying LLM-based solutions for data extraction.

CORE COMPETENCIES AND TECHNOLOGIES

Languages & Core:	Python (Advanced), R, SQL, Git, Docker
Machine Learning & AI:	Scikit-Learn, PyTorch, LangChain, Ollama, XGBoost, Transformers (Hugging Face)
Statistical Methods:	Causal Inference, Time-Series, Bayesian Statistics, Multivariate Analysis.
Cloud Services:	Google Cloud Platform (GCP), AWS
Visualization:	Tableau, Microsoft Power BI, Google Looker
Skills and Area of Expertise:	Research, Data Science, Big Data, Machine Learning (ML), Deep Learning (DL)

WORKING EXPERIENCE

Matana University - Tangerang, Indonesia	Full-Time (On-Site)	
• Head of the Statistics Undergraduate Programme		Mar 2025 – Present
Role and Key achievements:		
- Modernized the Statistics curriculum to align with Industry 4.0 standards, integrating Python, Deep Learning, and Cloud Computing into core coursework. - Directed undergraduate research groups focusing on Applied ML, resulting in implementation of algorithms for real-world datasets. - Led data-driven strategic planning for the department, utilizing internal data analytics to improve student retention and resource allocation.		
• Lecturer of the Statistics Undergraduate Programme		Aug 2024 – Present
Role and Key achievements:		
- Prepared and presented interesting and thorough lecture materials on the topics of data science, statistics, machine learning, or deep learning and used the latest data science methods to produce practical examples and assignments. - Led students in the process of conducting research projects to improve their AI/ML algorithms and data-driven decision-making skills. - Courses: - Algorithm and Programming in R and Python - Database and Data Retrieval in SQL with PostgreSQL and non-RDBMS with MongoDB - Database and Data Retrieval in Big Data (HDFS, Spark, Kubernetes, Kafka, Google Query, Heroku, etc) - Data Exploration and Data Visualization - Introduction to Data Science - Multivariate Analysis (CCA, Conjoint Analysis, Path Analysis, Factor Analysis, SEM, MDS, PSM, IV, etc.) - Numerical Methods - Profession Ethic in Statistics and Data Science - Statistical Computation		
Generative AI & LLM Engineer	Independent Consultant	May 2021 – Present
Role and Key achievements:		
High-Performance Inference Engineering: Engineered a production-grade LLM serving architecture using vLLM and 4-bit quantization (TGI), achieving a 15x increase in concurrent user capacity and a 40% reduction in cloud infrastructure costs by optimizing GPU throughput and reducing Time-To-First-Token (TTFT) to <200ms. Distributed Training Infrastructure: Architected a "scale-to-zero" ephemeral compute cluster integrating AWS Spot Instances, Ray, and Slurm, enabling cost-effective distributed training workflows that reduced GPU idle costs by ~30% while ensuring fault tolerance against node preemptions. Hybrid-Cloud Storage Optimization: Solved critical "Starving GPU" bottlenecks in deep learning pipelines by implementing a transparent POSIX-compatible caching layer (JuiceFS), bridging the gap between cost-effective S3 object storage and high-performance local NVMe compute.		

- **AutoML System Development:** Developed a distributed **Neural Architecture Search (NAS)** engine using **Ray Tune** and **Optuna**, replacing manual hyperparameter tuning with an automated, cost-aware system that optimizes model performance across distributed GPU nodes.
- **Secure GenAI Application Development:** Built privacy-first local analytics platforms using **Llama-3** and **PandasAI**, enabling non-technical stakeholders to query sensitive financial data via natural language, reducing manual reporting and review cycles by **40%**.

Outlier AI - San Francisco, USA | Part-Time (Remote)

Software Engineering for AI Training Data (Python)

Jun 2024 – Nov 2024

Role and Key achievements:

- Conducted rigorous technical review and Reinforcement Learning from Human Feedback (RLHF) for rigorous Python code generation models.
- Refactored inefficient Python scripts generated by AI models, achieving up to **15%** reduction in computational complexity while maintaining accuracy.
- Designed edge-case unit tests to validate model logic and ensure compliance with enterprise coding standards.

Indonesian Directorate General of Taxes - Semarang, Indonesia | Internship (Onsite)

Data Analyst

Nov 2019 – Dec 2019

Role and Key achievements:

- Done the data entry and had the verification for many different financial revenue that was sure about the accuracy and reliability of the records.
- From the financial revenue data, the trends and patterns were found by the data analysis, and it helped with the process of making decisions based on the new information.
- Prepared extensive annual financial reports on the income, thus dealing with strategic planning and organizational transparency was no longer an issue.

Partai Amanat Nasional - Semarang, Indonesia | Freelance (Onsite)

Data Warehouse Administrator

May 2017 – Oct 2017

Role and Key achievements:

- Conducted very detailed data verification and also accepted the verification process of Party's member population data by KPUD in compliance with their regulations, guaranteeing the safety and security of the data.
- Designed and looked over a database that was organized, thus the data management and organizational needs were simplified and data were on the hand accessibly.
- Contributed to a great extent to the database of the Party by changing and correcting the members who were wrongly categorized henceforth this led to the high accuracy and reliability of the party's database for the electoral process.

Badan Pusat Statistik - Kendal, Indonesia | Internship (Onsite)

Surveyor

Feb 2017 – Mar 2017

Role and Key achievements:

- Participated and assisted in the survey on food commodities which collected accurate and precise data on themes such as the proper materials, corn, and the like for BPS.
- Facilitated the process of the analysis of food commodity data, which was instrumental in detecting the trends and the insights supporting decision-making, and forming good policies were vital.
- Participated in the spatial survey, more precisely, in the areas of maize cultivation, where they lent a hand in generating the geographic data which is the backbone of the planning and the development of the agricultural sector.

KEY ENGINEERING PROJECTS

High-Throughput LLM Inference Architecture

- **Project:** The Inference Engine (LLM Serving Optimization)
- **Goal:** To solve the critical "slow money" problem in Generative AI by minimizing Time-To-First-Token (TTFT) and maximizing throughput on limited GPU resources.
- **Tech Stack:** Python, vLLM (PagedAttention), HuggingFace TGI, quantization (bitsandbytes), Prometheus, Grafana, Locust.
- **Impact:**
 - **Up-to 15x Capacity Increase:** Optimized serving architecture to handle ~150 concurrent users per GPU compared to <10 on baseline.
 - **Up-to 40% Cost Reduction:** Reduced projected infrastructure costs by ~\$800k/year by switching from naive pipelines to 4-bit quantized TGI and continuous batching.
 - **Latency Elimination:** Achieved <200ms TTFT targets under heavy load, validated through distributed load testing with Locust.

Hybrid-Cloud Storage Acceleration for Deep Learning

- **Project:** The Hybrid-Cloud Data Bridge
- **Goal:** Eliminating the "Starving GPU" bottleneck caused by network latency when training models on on-premise hardware using cloud-stored datasets (S3).
- **Tech Stack:** Docker, PyTorch, MinIO (S3), JuiceFS (POSIX Caching), Redis, NVIDIA Container Toolkit.
- **Impact:**
 - **I/O Saturation:** Implemented a transparent POSIX-compatible caching layer that saturates GPU compute capacity, removing network I/O wait times.
 - **Zero-Code Refactoring:** Enabled PyTorch DataLoaders to access S3 objects as local files without modifying training scripts.
 - **Cost Efficiency:** Bridged the gap between cheap Object Storage (Cloud) and fast NVMe (Local), avoiding expensive cloud egress fees for repeated epochs.

Distributed Training Infrastructure (Slurm & Ray)

- **Project:** Mini-HPC Cluster
- **Goal:** Designing a production-grade, ephemeral compute cluster that bridges legacy HPC scheduling (Slurm) with modern AI-native orchestration (Ray).
- **Tech Stack:** AWS EC2 (Spot Instances), Slurm, Ray Cluster, Bash Scripting, MinIO.
- **Impact:**
 - **Up-to 30% Cost Savings:** Architected a stateless "scale-to-zero" infrastructure leveraging AWS Spot Instances for heavy training jobs.
 - **Hybrid Orchestration:** Demonstrated versatility by implementing both Slurm (for monolithic batch jobs) and Ray (for elastic, granular actor scheduling) in a single environment.
 - **Fault Tolerance:** Built automated recovery workflows where compute nodes are treated as ephemeral "cattle," decoupled from the storage layer.

Local GenAI Analytics Platform

- **Project:** Secure Data Analysis with Llama 3 & PandasAI
- **Goal:** Democratizing advanced data analysis for non-technical users while maintaining data privacy by running LLMs locally (off-cloud).
- **Tech Stack:** Streamlit, PandasAI, Ollama (Local Llama 3), Python.
- **Impact:**
 - **Privacy-First AI:** Enabled sensitive financial/operational data analysis without sending datasets to external APIs (OpenAI/Anthropic).
 - **Natural Language Interface:** Reduced time-to-insight by allowing users to query CSV/Excel datasets using plain English instead of SQL or Python. Reducing manual review time by **40%**

Enterprise AutoML & Neural Architecture Search (NAS) System

- **Project:** Distributed AutoML Orchestration Engine
- **Goal:** To engineer a production-grade, fault-tolerant NAS system capable of discovering optimal neural architectures across distributed GPU clusters while strictly minimizing cloud compute costs.
- **Tech Stack:** Ray Tune, PyTorch, Optuna (TPE), AWS Spot Instances, MLflow, Docker.
- **Impact:**
 - **Up-to 40% Cost Reduction:** Achieved comparable model performance at **<10% of the cost** of traditional grid search (\$5 vs \$52) by leveraging Spot Instances and ASHA early-stopping algorithms.
 - **Resilient Architecture:** Designed a "Split-Plane" architecture that isolates control logic from compute, enabling **zero-data-loss recovery** during Spot Instance preemptions or OOM failures.
 - **Scalable Orchestration:** Implemented a distributed search engine using Ray Tune that scales linearly from a local workstation to a multi-node GPU cluster without code modification.

EDUCATION

Institut Teknologi Sepuluh Nopember - Surabaya, Indonesia

Master's Degree of Statistics

Jan 2022 – Apr 2024

- Demonstrated proficiency in conducting research by devising an innovative approach for scaling a previously thought non-scalable community detection algorithm, [published](#) in a peer-reviewed journal.
- Excelled in academic project, demonstrating skills in algorithm design, data structures, and machine learning.
- Completed highly demanding coursework in Statistical Process Control, Econometrics, Inference Statistics, Meta Analysis, Data Analysis, and Intensive Computational Statistics.

Universitas Muhammadiyah - Semarang, Indonesia

Bachelor's Degree of Statistics

Sep 2019 – May 2021

- Actively participated in statistical seminars, workshops, and conferences, enhancing both theoretical knowledge and practical skills in the field, with a focus on Python and R.
- Excelled in academic projects, demonstrating skills in time series analysis and exploratory data analysis, including techniques requiring graphical visualization for insightful data interpretation and presentation.
- Completed highly demanding coursework in Regression Analysis, Multivariate Analysis, Time Series Analysis, Categorical Data Analysis, and Financial Statistics.

Akademi Statistika Muhammadiyah Semarang - Semarang, Indonesia

Diploma III or Associate's Degree of Statistics

Sep 2016 – Sep 2019

- Actively participated in student organizations and volunteering initiatives, demonstrating leadership skills and a commitment to community engagement through statistical applications.
- Excelled in academic project, demonstrating skills in data mining, data structures, and their implementation using SQL, including proficient querying, database design, and optimization techniques for efficient data retrieval and manipulation.
- Completed highly demanding coursework in Probability Theory, Stochastic Processes, Bayesian Methods, Experimental Design, Calculus, Linear Algebra, Inventory Statistics, and Data Mining.

RESEARCH AND PUBLICATIONS

Google Scholar: [Amanatullah Pandu Zenklinov](#)

• Stunting Risk Factor Analysis Using Seemingly Unrelated Regression (SUR) Integrated with Machine Learning	2025
• The Fuzzy-Possibilistic Product Partition c-Means (FPPPCM) algorithm for Clustering the Welfare Levels of Regencies in East Java	2025
• Utilization of Flora in the School Environment as a Drawing Medium for Reading Children's Creativity	2025
• Intrusion Detection Systems (IDSs) using Multivariate Control Chart Hotelling's T2 with Dimensional Reduction of Factorial Analysis of Mixed Data (FAMD) and Autoencoder	2024
• Spatial Temporal Density Based Spatial Cluster of Application with Noise untuk Mengelompokkan Curah Hujan di Jawa Timur	2024
• Pengelompokkan Kemiskinan Kabupaten/Kota Di Jawa Tengah Menggunakan Algoritma Fuzzy Possibilistic C-Means dan Possibilistic Fuzzy C-Means	2021

COURSE AND CERTIFICATIONS

• Generative AI Engineering with LLMs Specialization	IBM	Apr 2025
• English Level C2 Advanced according to the Common European Framework of Reference (CEFR)	EF SET	Apr 2025
• English Level C1 Advanced according to the Common European Framework of Reference (CEFR)	EF SET	Jul 2024
• Introduction to Machine Learning on AWS	AWS	Jul 2024
• GenAI Chatbots: Create and Deploy OpenAI-Powered Chatbots	Coursera	Jun 2024
• Google Cloud Big Data and Machine Learning Fundamentals	Google	May 2024
• IBM Machine Learning Professional Certificate	IBM	Apr 2024
• Prompt Engineering	IBM	Mar 2024